

SYLLABUS MATH 289, FALL 2026

INSTRUCTOR

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SCHEDULE

Lectures are on Wednesdays from 4 pm to 5 pm in 3356 Mason Hall, starting on 9.2.26 and ending on 12.9.26 included. There won't be lecture on Wednesday 11.25.26.

There are weekly homework due on Wednesdays at 3pm. The first homework is due on 9.9. All the relevant information, material, and announcements are available at the course website: <https://umich.instructure.com/courses/858839>.

DESCRIPTION

One of the better ways to develop mathematical abilities is by solving problems using a variety of methods. Familiarity with numerous methods is a great asset to the developing student of mathematics. Methods learned in attacking a specific problem frequently find application in many other areas of mathematics. In many instances an interest in and appreciation of mathematics is better developed by solving problems than by hearing formal lectures on specific topics. **This course is intended for students who are enthusiastic about doing mathematics and solving challenging problems.** It is not restricted to honors students. This course is excellent preparation for the Putnam competition. The Putnam competition takes place on the first Saturday in December, and consists of two three-hour sittings separated by a lunch break. Each one consists of twelve challenging problems. The problems cover a range of advanced material in undergraduate mathematics, including concepts from group theory, set theory, graph theory, lattice theory, and number theory. The competition is considered to be very difficult: it is typically attempted by students specializing in mathematics, but the median score is usually zero or one point out of 120 possible, and there have been only five perfect scores as of 2021. In 2003, of the 3,615 students competing, 1,024 (28%) scored 10 or more points, and 42 points was sufficient to make the top percentile. More information about the competition are available at the [wikipedia page](#).

POLICY ON MOBILE PHONES, LAPTOPS, AND TABLETS

The use of mobile phones, laptops, and smartwatches is not allowed during the lecture, while the use of tablets is only permitted for taking notes. I will invite the students using their phones, laptops, smartwatches, or using a tablet not for taking notes of the 289 lecture to leave the classroom in the interest of the concentration of the class. Repeated misuse of electronic devices during the classes will cause a penalization up to a full letter grade in the final grade.

ATTENDANCE

You are required to come to class. Students who do not attend class seldom succeed in this course. You are allowed up to 2 unexcused absences, but any unexcused absences thereafter will result in a 1/3 of a letter grade drop from your final grade (e.g, an A- drops to a B+, etc), and these deductions stack with no cap. If you must miss class for any reason, it is your responsibility to get notes from a classmate and catch up on the material. You are also responsible for the information given in any announcements made during class.

HOMEWORK

Each week I will propose a number of problems during lecture. At the end of the lecture, I will select two of these problems as that week's homework assignment. You must submit your solutions to the two selected problems via Gradescope by 3pm on the following Wednesday, one hour before lecture. Late homework will not get any credit. For a problem to receive full credit, you must show all your work, explain your work as necessary, and present your work in a clean and clear format that is easily understood. You may collaborate with other students in solving the homework problems, but you must turn in a write-up in your own words. You may not post homework problems to internet discussion boards.

GRADING POLICY

Late homework (or not knowing the exercises assigned) will not get credits and will get a score of zero. Homework grades count 100% of the final grade. Your final grade depends on the final number of points you have. The total number of points you can get is 100, and the correspondence points-letter grade is given by the Table 1:

Points	Grade
≥ 95	A+
≥ 90	A
≥ 85	A-
≥ 80	B+
≥ 75	B
≥ 70	B-
≥ 65	C+
≥ 60	C
≥ 55	C-
≥ 50	D+
≥ 45	D
≥ 40	D-
≤ 39	E

TABLE 1. Grading Scale

The points are computed as follows. 100 points come from the HM assignments. After making the points uniform for each assignment by the formula (your points for one assignment)/(total number of points of that assignment) I will take the sum of the HM scores and divided it by the total number of HM. Finally, I multiply this number by 100. Note that some of you have regrade requests for various HM so the grade on canvas might not be updated. Before computing your final grade I will update all the HM scores from gradescope. I will let you know when they are final so you can double check all my computations.

OFFICE HOURS

Office hours are on Mondays and Tuesdays from 5:30pm to 6:20pm in my office.

EQUAL OPPORTUNITY

The University of Michigan is committed to providing equal opportunity for participation in all classes, programs, services and activities. Requests for accommodations by persons with disabilities may be made by contacting the Services for Students with Disabilities (SSD) Office located at G664 Haven Hall. The SSD phone number is 734-763-3000 and their website is ssd.umich.edu. Once your eligibility for an accommodation has been determined, please share this information with me at the beginning of the term, or at least two weeks before you need the accommodation for an exam, project, etc.